CHURN AND RETENTION DATA DOCUMENTATION

1. Purpose of This Document

This document provides a clear and careful explanation of how **churn** and **retention** data is defined, prepared, analyzed, and interpreted. It is intended for use in dashboards, reports, academic projects, and business reviews, ensuring that stakeholders understand *what the metrics mean*, *how they are calculated*, and *how to act on them*.

2. Business Context

In subscription-based businesses (telecom, SaaS, OTT, utilities, etc.), revenue depends on retaining customers over time. Understanding churn and retention helps to:

- Reduce customer loss
- Improve customer lifetime value (CLV)
- Identify at-risk customers early
- Design effective retention strategies

3. Key Definitions

3.1 Customer Churn

Customer Churn refers to customers who stop using the service during a given time period.

- **Voluntary churn**: Customer cancels service by choice
- **Involuntary churn**: Service stops due to non-payment, technical issues, or policy violations

Business Impact: High churn directly reduces revenue and increases acquisition costs.

3.2 Customer Retention

Customer Retention refers to customers who continue using the service over a given time period.

Retention focuses on stability, loyalty, and long-term value.

4. Dataset Overview (Typical Structure)

Category	Example Fields
Customer Info	Customer ID, Age, Gender, Location
Subscription	Plan Type, Contract Duration, Start Date
Usage	Call Minutes, Data Usage, Login Frequency
Billing	Monthly Charges, Total Charges, Payment Method
Support	Complaints, Tickets Raised
Status	Churn (Yes/No)

5. Data Preparation & Cleaning Steps

5.1 Data Quality Checks

- Remove duplicate Customer IDs
- Handle missing values (mean/median/imputation)
- Validate date formats
- Standardize categorical values (Yes/No, Male/Female)

5.2 Outlier Treatment

- Identify extreme values in usage and billing
- Validate whether outliers are realistic
- Cap or remove invalid values

5.3 Feature Engineering

- Customer Tenure (months)
- Average Revenue Per User (ARPU)
- Usage per month
- Complaint frequency

6. Churn Metrics & Calculations

6.1 Churn Rate

Definition: Percentage of customers lost during a period

Formula:

Churn Rate = (Number of Churned Customers / Total Customers at Start) × 100

6.2 Retention Rate

Definition: Percentage of customers retained during a period

Formula:

Retention Rate = 100 − Churn Rate

6.3 Customer Lifetime (Tenure)

Definition: Average duration a customer stays active

Formula:

Customer Lifetime = Average number of months before churn

7. Segmentation Analysis

Churn should be analyzed across different dimensions:

- Contract Type (Monthly vs Long-term)
- Plan Type (Basic / Premium)
- Payment Method
- Geography
- Usage Levels
- Support Interaction Count

Goal: Identify high-risk segments.

8. Retention Driver Analysis

Key factors influencing retention typically include:

- Contract duration (longer contracts → lower churn)
- Service usage (higher engagement → lower churn)
- Pricing sensitivity
- Quality of customer support
- Payment convenience

Correlation and comparative analysis are used to validate drivers.

9. Dashboard KPIs (Recommended)

Core KPIs

- Total Customers
- Churned Customers
- Churn Rate (%)
- Retention Rate (%)
- Average Customer Tenure
- Monthly Revenue

Visuals

- Churn by Contract Type (Bar Chart)
- Churn Trend Over Time (Line Chart)
- Retention by Plan Type (Stacked Chart)
- Customer Lifetime Distribution (Histogram)

10. Interpretation Guidelines

- A **high churn rate** indicates dissatisfaction or pricing issues
- A **low churn but low revenue** may indicate underpricing
- Sudden churn spikes usually signal operational issues
- Long-tenure customers are the most valuable

Always interpret metrics in combination, not isolation.

11. Business Recommendations Framework

Based on insights:

- Introduce loyalty rewards for long-tenure customers
- Improve onboarding for new customers
- Offer contract upgrades with incentives

- Target high-risk segments with proactive support

12. Assumptions & Limitations

- Churn definition depends on business rules
- Historical data may not predict sudden market changes
- External factors (competition, regulation) are not included

13. Conclusion

Churn and retention analysis is a critical component of sustainable business growth. Proper data preparation, consistent metric definitions, and thoughtful interpretation enable actionable insights that directly improve customer experience and profitability.